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DISTRIBUTION UTILITY MEET DUM 2026

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DUM 2026 CONFERENCE SESSION

India Energy Stack (IES)

Connectivity as a foundation for IES

Presented By

Teppo Hemiä, CEO, Wirepas

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in

SMART METERING REQUIREMENTS ARE EVOLVING

DISTRIBUTION UTILITY MEET DUM 2026

ISGF India Smart Grid Forum

Analogue/AMR

- Billing
- Reading app



- Flat rate
- Monthly reads
- Manual / AMR / Drive by
- Wireless M-Bus

1st Gen Smart

- Billing
- MDM
- HES



- Time of Use
- Hourly reads
- PLC, RF Mesh, NB-IoT, LTE-M

2nd Gen Smart

- Billing
- MDM
- HES



- Time of Use, Dynamic Pricing
- 15 min reads
- RF Mesh, 4G

RDSS/SBD

+

3rd Gen Smart

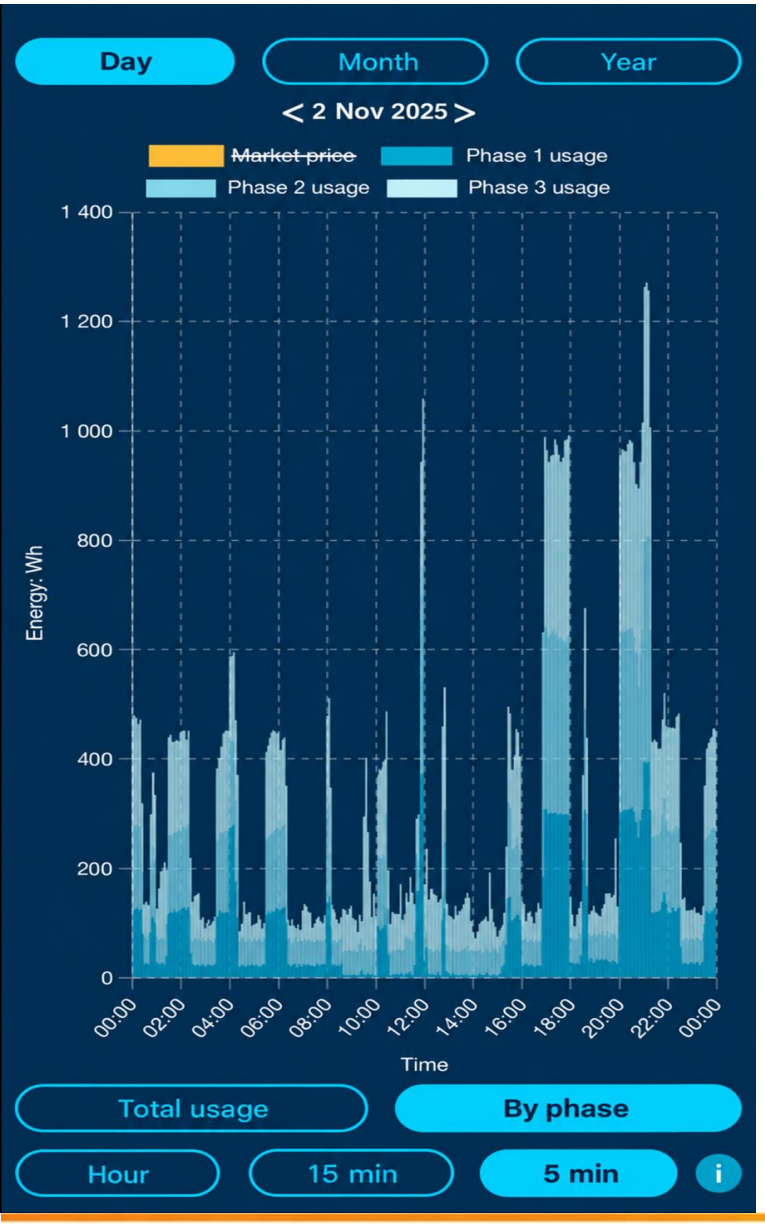
- Apps, CIU
- Load monitor and control
- Edge compute



- Real-Time Pricing
- 5 min to < 1s reads
- Event stream
- NR+, 5G
- Data as a Service

Data as a Service

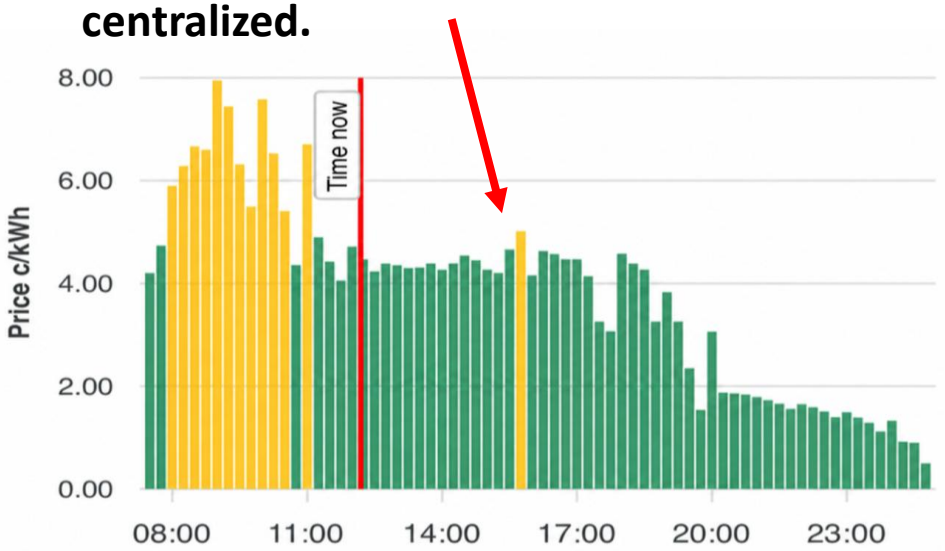
EXAMPLE: FINLAND



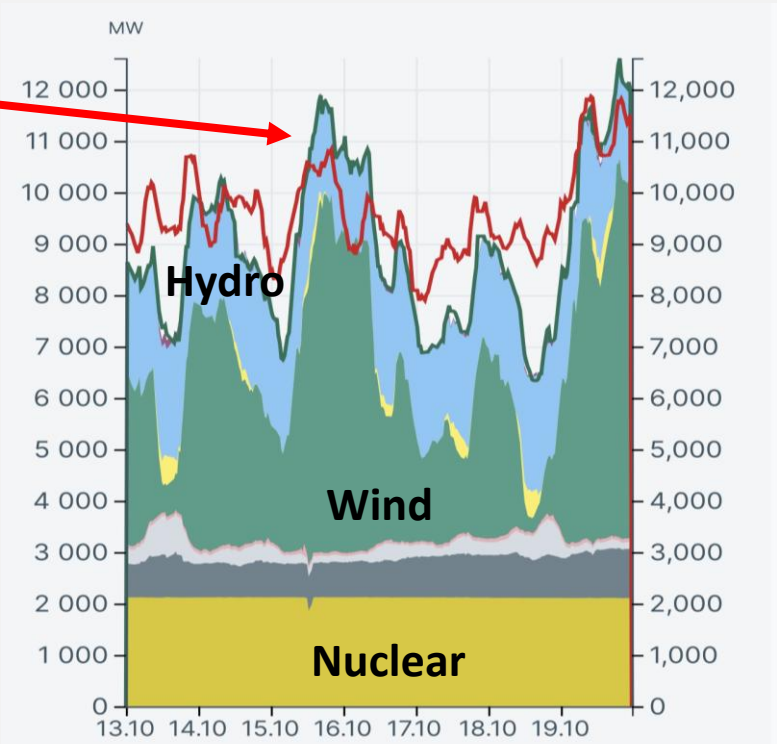
In Finland, 2nd - 3rd generation of meters, **5min reading interval**, three phases. Consumer see it almost real time.

Public site for production type and amount for the country. **Renewables impact**.

Dynamic prices are set daily in an auction with 15min interval, 24h ahead. Both, buying and selling. E.g. excess of solar power. **Still centralized.**



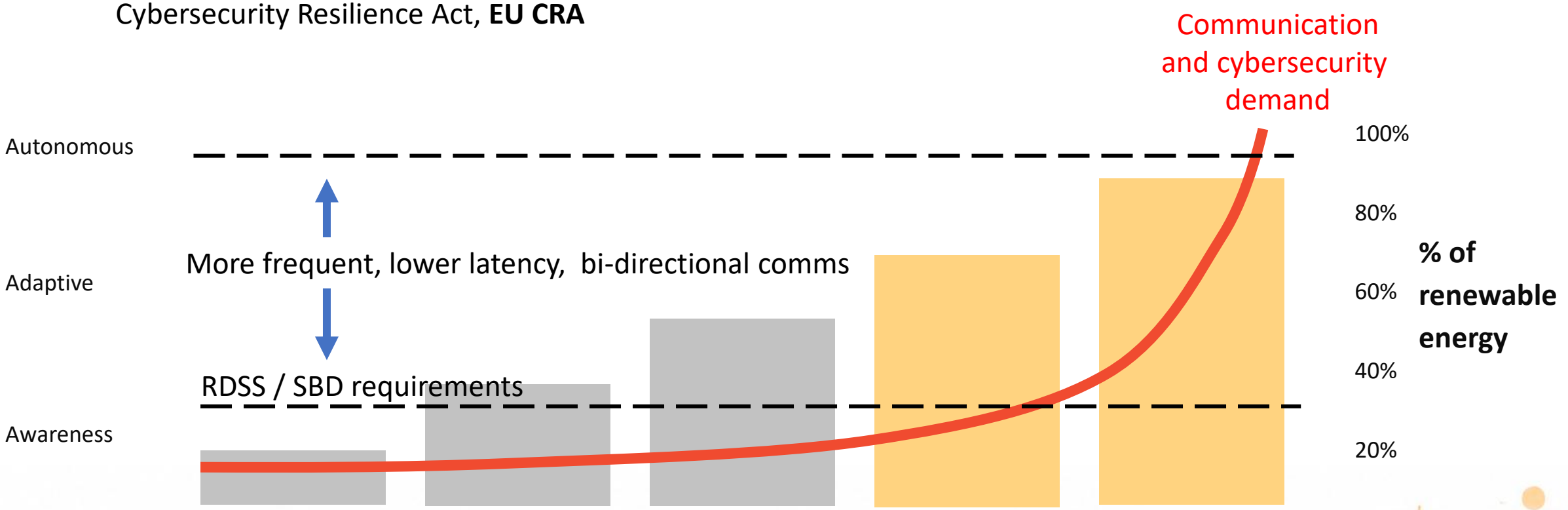
Finland is part of Nord Pool



Production type

IES' and Renewables' mutual impact on connectivity

- IES as fully decentralized/autonomous is very ambitious!
- IES + renewables, dynamic pricing and grid balancing drives connectivity
- Standardization is key to level the whole industry
  - Europe: European Telecommunications Standards Institute, **ETSI** European Union Cybersecurity Resilience Act, **EU CRA**



Conceptual picture

EUROPEAN AND INDIAN STANDARDS ECOSYSTEM.

ETSI, BIS, TSDSI builds strategic view for the future.

NR+ by ETSI is now part of TSDSI. Major step for AMI as it is connectivity that is order of magnitude more performance than what RDSS/SBD set today

Wirepas Mesh is NR+ based solution
Wirepas Mesh is EU CRA compliant that is well aligned with CERT-In.

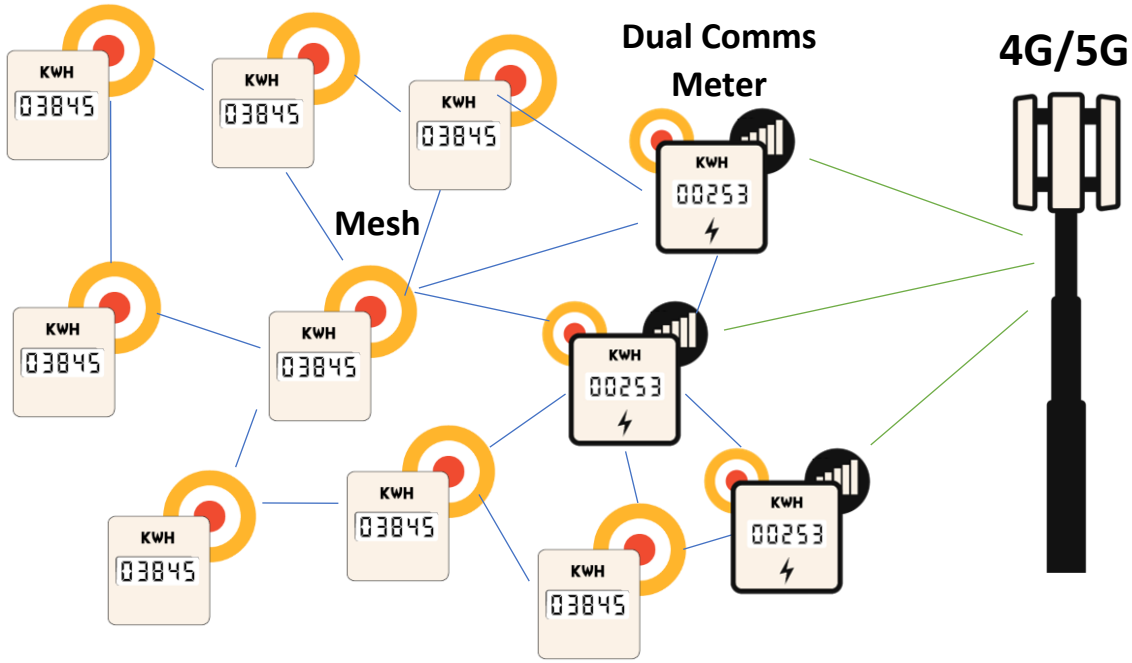
Wirepas is committed to be **the first to enable Indian chipsets.**



USE OF NR+ WITH CELLULAR, BOTH EVOLVES

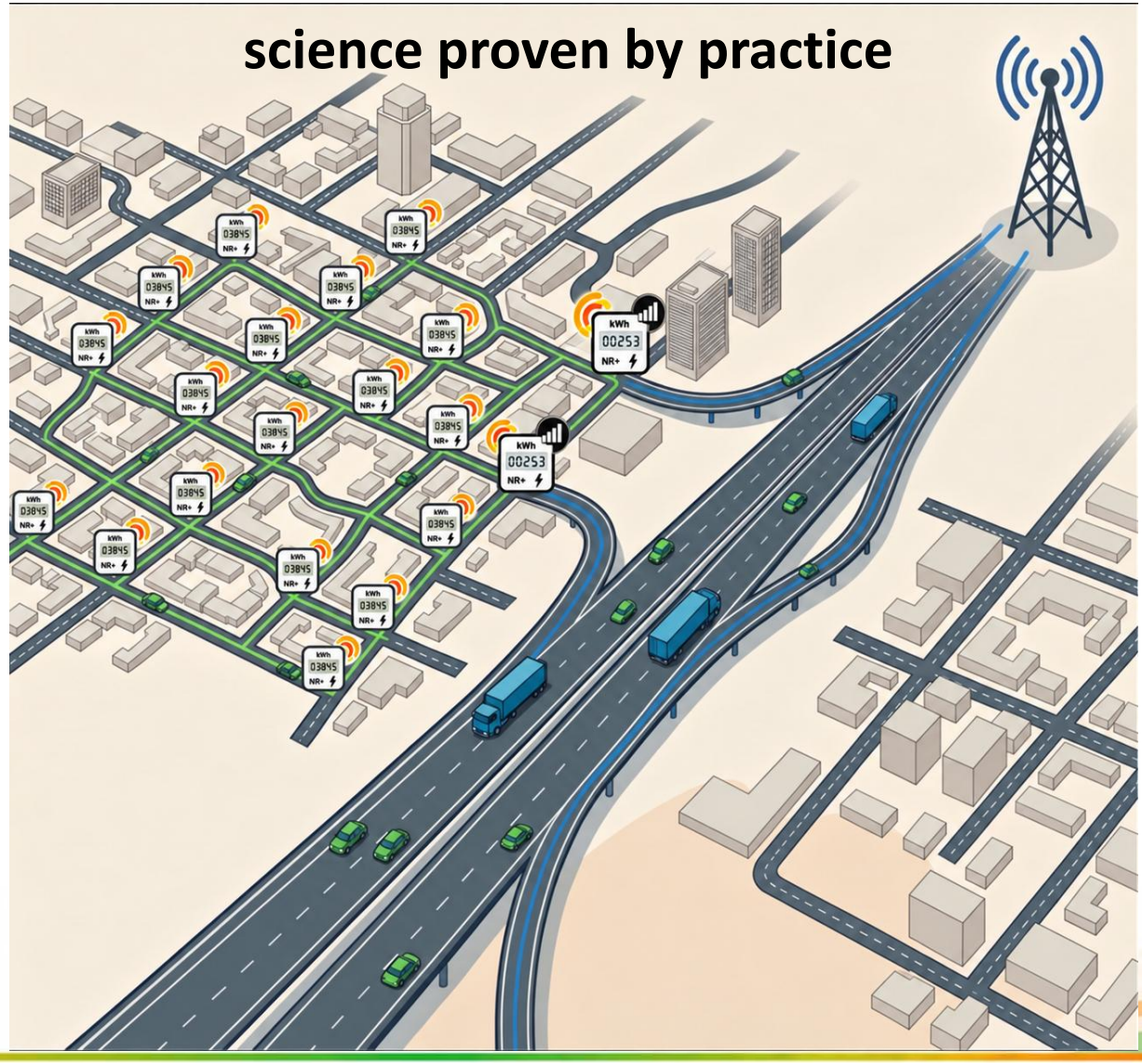
Leverage the best from both technologies

- 4G provides wide area coverage and high throughput **like a highway**
- Mesh provides local coverage with local agility to connects smart meters via multi hop mesh network **like local streets**

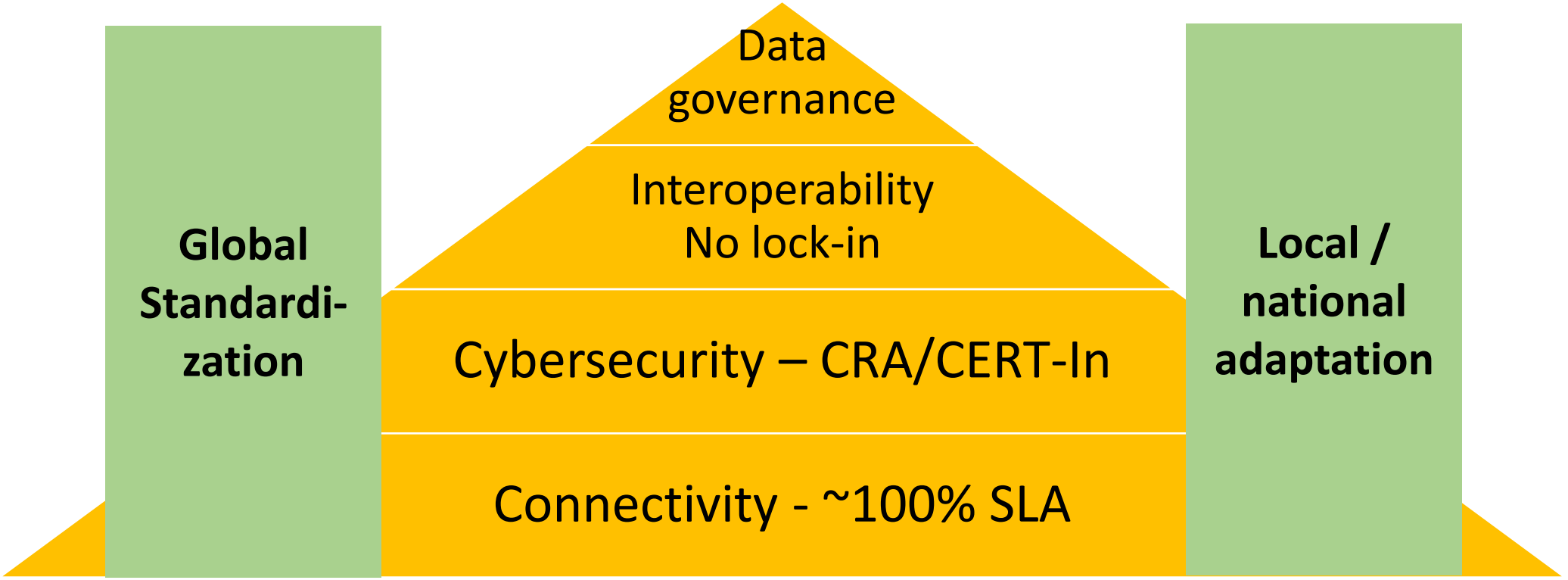


Local autonomous self healing network
Multi route, automatic reconfiguration.

Wide area coverage
High throughput low latency



IES NEEDS TRUSTED DATA FROM THE GRID EDGE



Dilemma: AMI procurement today determines how much freedom utilities will have to adopt new applications tomorrow.

THANK YOU

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